

BOURGES AP, France

WMO: 072550

Lat: 47.0592N

Lon: 2.3597E

Elev: 161

StdP: 99.40

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-3.2	-8.9	1.8	-3.0	-6.6	2.2	-1.5	11.9	10.6	10.5	9.8	3.1	80	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.5	20.4	30.2	19.7	28.2	19.0	21.6	29.5	20.7	28.1	19.8	26.4	3.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	24.1	18.2	13.4	23.0	17.5	12.8	22.3	63.5	29.8	60.2	28.2	57.3	26.4	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.0	7.0	-7.9	35.6	2.2	2.1	-9.5	37.2	-10.8	38.4	-12.0	39.6	-13.6	41.2		
			DB	WB	-8.4	23.3	2.2	1.1	-9.9	24.0	-11.2	24.7	-12.5	25.3	-14.1	26.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	4.4	5.1	8.3	10.9	14.7	18.5	20.5	20.2	16.5	13.0	7.7	4.9	(6)
(7)		DBStd	6.77	4.01	3.82	3.44	3.31	3.32	3.48	3.23	3.33	3.06	3.64	3.57	4.03	(7)
(8)		HDD10.0	685	177	140	74	27	3	0	0	0	15	86	162		(8)
(9)		HDD18.3	2508	433	371	313	223	121	41	14	16	72	168	320	417	(9)
(10)		CDD10.0	1447	2	2	21	54	148	254	325	318	196	107	16	4	(10)
(11)		CDD18.3	228	0	0	0	1	8	44	81	75	17	2	0	0	(11)
(12)		CDH23.3	2296	0	0	0	17	97	452	798	747	167	18	0	0	(12)
(13)		CDH26.7	801	0	0	0	1	13	151	293	301	40	2	0	0	(13)
(14)	Wind	WSAvg	3.4	3.8	3.8	3.7	3.5	3.2	3.1	3.1	2.9	3.0	3.2	3.4	3.8	(14)
(15)	Precipitation	PrecAvg	733	60	55	57	57	79	58	59	58	60	65	65	64	(15)
(16)		PrecMax	1009	147	139	139	182	175	127	177	163	179	187	149	147	(16)
(17)		PrecMin	497	5	6	4	7	15	0	7	4	8	4	20	7	(17)
(18)		PrecStd	122	31	29	34	34	37	29	36	38	39	38	28	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	17.1	21.6	25.5	28.4	33.8	34.7	36.9	30.5	25.9	18.1	14.5	(19)
(20)		0.4%	MCWB	11.3	11.1	12.9	16.1	19.5	21.4	20.7	21.2	18.1	17.6	14.2	12.0	(20)
(21)		2%	DB	12.5	14.0	18.8	22.5	26.0	30.4	32.1	32.7	27.5	22.9	15.6	13.0	(21)
(22)		2%	MCWB	10.8	9.9	11.8	14.4	17.4	20.1	20.5	20.2	18.0	16.9	13.1	11.2	(22)
(23)		5%	DB	11.2	12.0	16.4	20.0	23.9	28.0	29.9	29.9	25.1	20.5	14.2	11.9	(23)
(24)		5%	MCWB	9.7	9.2	10.8	13.1	16.8	19.3	19.6	19.8	17.2	15.9	12.2	10.4	(24)
(25)		10%	DB	10.0	10.6	14.3	17.5	21.7	25.8	27.8	27.4	22.9	18.4	12.9	10.7	(25)
(26)		10%	MCWB	8.6	8.6	10.0	12.0	15.6	18.4	18.8	18.8	16.4	15.0	11.2	9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.0	14.1	17.0	20.4	22.8	22.4	22.6	20.6	18.7	15.0	12.7	(27)
(28)		0.4%	MCDWB	13.0	15.2	19.0	24.3	26.6	30.5	30.8	30.5	26.5	23.3	16.3	13.7	(28)
(29)		2%	WB	11.1	10.7	12.7	15.2	18.7	21.3	21.3	21.4	19.2	17.4	13.6	11.6	(29)
(30)		2%	MCDWB	12.3	12.8	16.7	21.2	24.3	28.6	29.5	29.5	24.7	21.3	15.3	12.7	(30)
(31)		5%	WB	9.9	9.7	11.6	13.8	17.3	20.1	20.4	20.5	18.1	16.5	12.4	10.6	(31)
(32)		5%	MCDWB	11.0	11.6	15.1	18.6	22.5	26.2	27.9	28.0	23.0	19.9	13.9	11.7	(32)
(33)		10%	WB	8.7	8.7	10.6	12.5	16.1	18.9	19.6	19.5	17.2	15.4	11.3	9.5	(33)
(34)		10%	MCDWB	9.8	10.5	13.8	16.6	20.7	24.5	26.2	26.0	21.7	18.1	12.7	10.6	(34)
(35)	Mean Daily Temperature Range	MDBR	5.6	7.3	9.1	10.0	10.0	10.4	10.8	10.9	10.4	8.7	6.4	5.7	(35)	
(36)		5% DB	MCDBR	6.6	10.0	13.3	14.6	13.9	14.5	14.7	15.3	14.4	11.6	8.0	6.7	(36)
(37)		5% DB	MCWBR	5.1	6.8	7.5	7.7	7.0	6.0	5.4	5.5	6.3	6.4	5.6	5.4	(37)
(38)		5% WB	MCDBR	6.0	8.7	11.0	13.1	12.4	12.9	13.0	13.1	11.4	10.4	7.4	6.2	(38)
(39)		5% WB	MCWBR	5.0	6.3	6.7	7.2	6.6	5.8	5.1	5.3	5.6	6.0	5.5	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.336	0.364	0.396	0.399	0.410	0.404	0.396	0.364	0.358	0.329	0.313	(40)	
(41)		taud	2.456	2.399	2.327	2.236	2.277	2.283	2.295	2.334	2.404	2.428	2.456	2.445	(41)	
(42)		Ebn at Noon	772	828	854	857	866	856	855	848	845	790	746	727	(42)	
(43)		Edh at Noon	67	87	108	129	129	129	126	116	99	83	67	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.05	1.88	3.13	4.41	5.18	5.94	5.74	5.03	3.88	2.38	1.28	0.89	(44)	
(45)		RadStd	0.14	0.28	0.38	0.54	0.49	0.56	0.42	0.45	0.29	0.18	0.14	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page